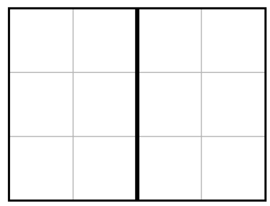
Halves and Fourths — Are the Parts Really Equal?

Some friends cut shapes into parts and gave the parts names. A part is a **half** only when the two parts hold the **same** number of squares. A part is a **fourth** only when all four parts hold the **same** number of squares.

- Count the little squares inside each part.
- If the counts are not the same, the parts are not equal shares.
- Then fix what the friend did wrong.

1. Warm up. Rosa cut this rectangle into two parts. Count the squares in each part.

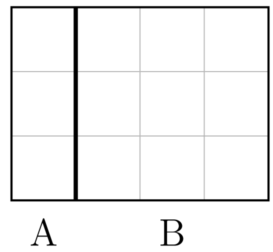


A B

Are Part A and Part B the same size? Write = if they are the same, or write < or > to show which one is bigger.

Answer: ________________

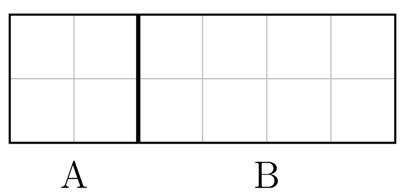
2. Find and fix. Kim cut this rectangle and said, "Part A is one half." Part A covers 3 squares. Part B covers 9 squares.



Write =, <, or > between the two counts to show what is really true. Then tell a friend why Part A is not one half.

Answer: ________________

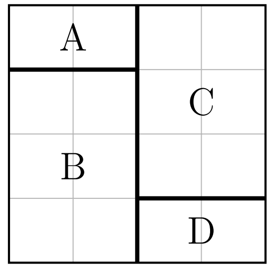
3. Find and fix. Ben cut this rectangle into two parts. Part A has 4 squares and Part B has 8 squares. Ben wants both parts to be halves, so he says, "Move 4 squares from Part B over to Part A."



Ben's move does not work. How many squares should really move from Part B to Part A so that the two parts are halves?

Answer: ________________

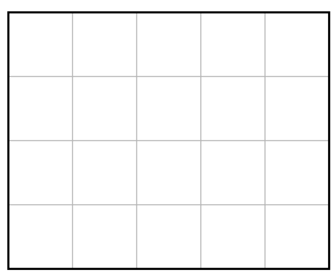
4. Kai cut this square into four pieces and called them fourths. Piece A covers 2 squares, Piece B covers 6 squares, Piece C covers 6 squares, and Piece D covers 2 squares.



Write the four counts in order, smallest first.

Answer: ______________

5. Find and fix. Ivy's rectangle has 20 little squares. She wants to cut it into 4 equal fourths. Ivy folded the rectangle one time, counted one of the two parts, and wrote 10 squares for a fourth.



Ivy's number is too big. How many squares should each fourth really have?

Answer: ______________

6. Challenge. Draw one straight line on this rectangle to cut it into two halves. Then write how the squares show that your two parts are halves.